Homework 3 (Convex and Piecewise Linear Functions)

February 3, 2026

Questions 1-4 are worth 10 points. Question 5 is worth 20 points.

1)

Let $f : \mathbb{R}^n \to \mathbb{R}^m$, and $g : \mathbb{R}^m \to \mathbb{R}^k$ both be affine. Show $g \circ f$ is also affine.

2)

Suppose $f, g : \mathbb{R} \to \mathbb{R}$ are both convex. Also suppose g is non-decreasing. Show that their composition, $g \circ f$ is also convex.

3)

Prove that the a function $f : \mathbb{R}^n \to \mathbb{R}$ is affine if and only if f is both convex and concave. (Hint: $f(\vec{x})$ is affine if and only if $g(\vec{x}) = f(\vec{x}) - f\left(\vec{0}\right)$ is linear).

4)

Let

$$f(\vec{x}) = \sum_{i=1}^{m} f_i(\vec{x})$$

where each $f_i : \mathbb{R}^n \to \mathbb{R}$ is convex.

a)

Show that $f(\vec{x})$ is convex.

b)

If each f_i is piecewise linear convex, show that $f(\vec{x})$ is piecewise linear convex.

5)

For each set, S, determine if S is convex. If S is convex, prove this. If S is not convex, explain why.

a)

The unit disk:

$$S = \left\{(x, y) \,|x^2 + y^2 \leq 1\right\}$$

b)

Let $\vec{x}_1, \vec{x}_2, \ldots, \vec{x}_n$ be a collection of n points in $\mathbb{R}^k$. Let $C = \{\vec{c} \in \mathbb{R}^n | c_1 + c_2 + \ldots + c_n = 1,\ 0 \leq c_1, c_2, \ldots, c_n \leq 1\}$. Suppose $n \geq 2$

$$S = \left\{ \sum_{i=1}^{n} c_i \vec{x}_i | \vec{c} \in C \right\}$$

c)

Let $\mathbb{R}^{n\times n}$ be the set of all $n \times n$ matrices. Suppose $n \geq 2$.

$$S = \left\{A \in \mathbb{R}^{n\times n} | \det(A) = 1\right\}$$

d)

Let $f : \mathbb{R}^n \to \mathbb{R}$ be a convex function and let c be some constant.

$$S = \{\vec{x} \in \mathbb{R}^n | f(\vec{x}) \leq c\}$$